

Maneesh Anand K N

Engineering Leader | Global Software Delivery | Cloud Platforms | Embedded Systems | Digital Health | AI/ML

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Engineering leader specializing in building and scaling global software organizations delivering cloud platforms, embedded systems, and regulated medical software. Proven track record of transforming engineering execution, modernizing development practices, and aligning Product, Architecture, and Engineering to deliver complex multi-product portfolios. Led globally distributed organizations of 60+ engineers, driving organizational change, improving delivery predictability, and embedding security and regulatory compliance into engineering operations. Recently expanded expertise in AI/ML, Generative AI, and production model deployment to accelerate software innovation and engineering productivity.

SKILLS & CORE CAPABILITIES

- **Engineering Leadership:** Organizational Design, Global Team Leadership, Engineering Strategy, Portfolio Leadership, Technology Roadmaps, Budget Ownership, Talent Development, Change Management (PROSCI), Danaher Business System (DBS), M&A Due Diligence & Integration, Innovation Management
- **Software Engineering:** SaaS / DaaS, Cloud Platforms (AWS, Azure), Embedded Systems, Platform Engineering, DevSecOps, SDLC Governance, IoT/IIoT Systems, Scalable Architecture
- **Execution Excellence:** Agile / SaaS Transformation, Portfolio & Program Governance, Release Management, Risk & Dependency Management, KPI Development, Vendor & Contractor Leadership, Process Maturity (CMMI/SEI-CMM L5, P-CMM, TL 9000)
- **Security & Compliance:** HIPAA, GDPR, IEC 62304, ISO 13485, ISO 27001, Secure SDLC, SBOM, FedRAMP, Incident Response, Vulnerability Management
- **AI / ML & Data:** Python, TensorFlow, Keras, Scikit-learn, Streamlit, Flask, Deep Learning (Neural Networks, CNNs, Transformers), NLP (LLMs, Prompt Engineering, RAG), Supervised & Unsupervised Learning (Regression, Trees, Clustering, Bagging, Boosting), Model Tuning & Evaluation, Data Analysis & Visualization (Pandas, NumPy, Seaborn), Hugging Face, Docker, Google Colab

PROFESSIONAL EXPERIENCE

Career Break & AI/ML Upskilling | 11/2023 – Current

- Completed an intensive postgraduate program in Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI at UT Austin, ranked 4th among a cohort of 200+ students.
- Designed and deployed production-ready AI/ML applications (Python, TensorFlow, Keras, Hugging Face, Docker, Streamlit, Flask) spanning predictive analytics, computer vision, recommendation systems, LLM/RAG applications, and intelligent automation.

Danaher / Beckman Coulter Diagnostics, Austin, TX

Senior Manager, Software R&D | 07/2021 – 11/2023

Led global Clinical Informatics R&D for a portfolio of SaaS/DaaS and regulated software products, spanning cloud, on-prem, and hybrid deployments across highly regulated healthcare environments. Managed a distributed organization of 60+ engineers and leaders across US, Europe and India, balancing delivery predictability, regulatory compliance, and platform evolution.

- Unified the engineering operating model by standardizing SDLC governance, tooling, and release processes, improving delivery consistency and reducing operational overhead.
- Aligned engineering teams across cloud, on-premise, and hybrid workstreams to deliver concurrent releases, improving delivery predictability and reducing last-mile risk.
- Partnered with Product and Architecture leadership to transform strategic priorities into executable engineering roadmaps, balancing capacity, sequencing, regulatory milestones, and customer commitments.
- Built a portfolio-wide execution framework that surfaced delivery risks early, accelerated executive decision-making, and strengthened cross-team accountability for on-time delivery.
- Modernized engineering quality practices — test strategy, release readiness criteria, and defect management — increasing engineering productivity by 20% and secure release throughput by 80% within 14 months.
- Embedded security, privacy, and regulatory compliance (HIPAA, GDPR, IEC 62304, ISO 13485, ISO 27001) into engineering delivery practices, achieving full compliance (audit pass) across products and vendors.

- Achieved 100% cybersecurity compliance across all Open Innovation (OI) partners within 2 months, a first in the business unit's history.
- Established a structured operating rhythm for execution governance, including status reporting, risk/issue reviews, and release readiness updates for executive leadership.
- Advanced DevSecOps maturity by operationalizing vulnerability management and incident response practices and SBOM, aligning with federal cybersecurity expectations under Executive Order 14028.
- Led FedRAMP gap assessment and cloud security enhancements, strengthening compliance posture and readiness for federal standards.
- Built a 24x7 CloudOps strategy supporting three business units, driving scalability and reducing annual operating costs by nearly \$200K.
- Improved cloud cost governance across AWS and Azure through engineering collaboration and operational visibility, reducing annual cloud spend by an additional 10%.
- Strengthened engineering culture through targeted initiatives that increased employee engagement participation by 23% within 6 months.
- Led organizational change, hiring, and leadership development initiatives, building high-performing global teams in a matrixed environment; recognized as the first certified Change Management Practitioner (PROSCI) under the senior VP reporting structure.
- Embedded Danaher Business System (DBS) practices into engineering operations by establishing KPIs, visual management and continuous improvement mechanisms across software delivery.
- Supported M&A integration readiness and technical due diligence efforts, aligning people, engineering processes, and tools for seamless integration.
- Evaluated HL7-based interoperability and medical device data integration capabilities of a potential M&A partner, assessing alignment with existing platform architecture and scalability requirements; provided technical due diligence inputs for acquisition decision-making.

Tech Environment: *Java, React, C++, .NET, C#, JavaScript, AWS, ADO/Azure, OCI, Snowflake, JIRA, Confluence, Bamboo, Jenkins, SharePoint, TFS, Angular, Adobe-XD, Figma, Linux, VPM/iObeya*

Viavi Solutions, Indianapolis, IN

Senior Manager, Software Engineering | 02/2016 – 07/2021

Led global engineering delivery for a portfolio of embedded software, DSP algorithms, firmware, cloud, analytics, and web platforms supporting communications test and measurement products. Managed a distributed organization of 60+ engineers across the US and Europe, driving engineering execution, organizational transformation, and product scalability. Provided strategic oversight on program delivery, risk mitigation, and cross-functional coordination across the product lifecycle using a People, Process, Tools framework.

- Directed delivery of 9 global products, aligning engineering execution with product strategy and revenue growth (\$77M annually) - delivered within a TL 9000 certified environment.
- Managed \$10M+ budgets across multiple business units, optimizing resource allocation and product lifecycle investment strategies.
- Established a portfolio-level planning and execution framework, improving organizational efficiency by 15%.
- Implemented technical debt and prioritization strategies, increasing engineering efficiency by 25%.
- Led Agile transformation across geographically distributed teams, applying structured change management models (Kotter's, Kübler-Ross) to drive Scrum adoption in a predominantly waterfall environment; improved integration with hardware teams and accelerated delivery timelines by 30%.
- Played a key role in building and scaling a Center of Excellence in Romania, strengthening global delivery capabilities and innovation.
- Enabled strong cross-functional collaboration across Product, Sales, Hardware, Manufacturing, and Operations.

Tech Environment: *C, C++, Qt, JQL, JIRA, JavaScript, Confluence, Bamboo, Jenkins, SharePoint, Angular, VxWorks, Linux, iOS, AWS*

Flextronics, Irving, TX

Consultant Agile Practitioner | 12/2015 – 02/2016

Drove Agile transformation in a regulated medical environment, establishing structured engineering execution frameworks and governance.

- Delivered the Agile Framework for two Fortune 500 medical device clients.
- Led rapid Agile adoption across cross-functional teams, achieving full integration within 2 months.
- Designed and implemented engineering program governance frameworks aligned with IEC 62304 and ISO 13485.
- Enabled alignment across software, hardware, QA, and regulatory teams for improved execution and compliance.

Tech Environment: C, C++, JIRA, Confluence, SharePoint

New York Air Brake (Train Dynamics Systems), Irving, TX

Software Engineering Manager | 11/2005 – 09/2015

Progressed from individual contributor to engineering leader within the Data Systems Group, managing embedded software, IoT, GIS and predictive analytics platforms for North American and global railroad customers.

- Led a 23-member engineering team delivering embedded software, machine learning, predictive analytics, and early IoT solutions supporting railroad safety and operations.
- Delivered 5 commercial software products, including the Data Systems platform, establishing a competitive foothold against established rail technology providers.
- Drove Scrum adoption and introduced Earned Value Management practices, strengthening program management rigor and delivery predictability across engineering initiatives.
- Partnered directly with customers to define product requirements & priorities, taking on product ownership responsibilities and providing cross-functional leadership across software development, QA, & data systems.
- Developed real-time tracking and asset management systems that improved railroad safety and operational efficiency.

Tech Environment: C, C++, Qt, Shell scripting, R, Perl, JQL, JIRA, Confluence, Jenkins, Bamboo, QNX RTOS, Linux, VMWare, AWS

ADDITIONAL RELEVANT EXPERIENCE

Wipro Technologies — Software Project Lead (Telecom — Lucent, Nortel) | 06/2001 – 08/2003

- Led telecommunications software development delivery for Lucent and Nortel within SEI-CMM Level 5 and People CMM (P-CMM) certified process environments.

Spectrum InfoTech — Project Lead (Aviation & Embedded Systems) | 03/1999 – 06/2001

- Delivered embedded aviation hardware & software while progressing into technical leadership.

EDUCATION

- Post Graduate Program in AI & Machine Learning – UT Austin
- MBA – University of Texas at Dallas
- MSEE – University of Texas at Arlington

CERTIFICATIONS

Certified Change Management Practitioner (Danaher/PROSCI) • Certified SAFe Agilist • Certified Scrum Master • Certified Scrum Product Owner

RECOGNITION & PUBLICATIONS

- "Scholar of Distinction" - School of Management at UT Dallas.
- Knorr-Bremse "Responsibility" Value Award
- 2nd place, co-authored paper "Neural Networks Applied to Periodic Table" - IEEE student's chapter.
- Publication, "Linear Prediction Applied to Periodic Table" - IEEE student's chapter.